

Quang Dao

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Education

- 2022–Present **Carnegie Mellon University, Pittsburgh, PA**
PhD in Computer Science. *Advisors:* Aayush Jain and Riad Wahby.
- 2020–2022 **University of Michigan, Ann Arbor, MI**
MA in Mathematics. *Advisor:* Paul Grubbs
- 2016–2020 **Columbia University, New York, NY**
BA in Mathematics and Computer Science

Research Interests

My research focuses on the **design**, **formal verification**, and **performance optimization** of **succinct zero-knowledge proofs (SNARKs)**. I currently work on building modular, machine-checked verification tools for cryptographic protocols, including SNARKs, in Lean (ArkLib, VCVio, PolyFun). I also design and optimize prover algorithms for SNARKs. My earlier work explored foundational post-quantum cryptography from code-based assumptions.

Publications

(* = first author(s))

8. Behzad Abdolmaleki, Matteo Campanelli, **Quang Dao**, Hamidreza Khoshakhlagh. On the Simulation-Extractability of Proof-Carrying Data. *PKC 2026*.
7. Carl Kwan*, **Quang Dao***, Justin Thaler. Verifying Jolt zkVM Lookup Semantics. *Financial Cryptography 2025*.
6. **Quang Dao**, Aayush Jain. Lossy Cryptography from Code-Based Assumptions. *CRYPTO 2024*. **Best Paper from Early Career Researchers**.
5. **Quang Dao**, Aayush Jain, Zhengzhong Jin. Non-Interactive Zero-Knowledge from LPN and MQ. *CRYPTO 2024*.
4. **Quang Dao**, Yuval Ishai, Aayush Jain, Huijia Lin. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN. *CRYPTO 2023*.
3. **Quang Dao***, Jim Miller*, Opal Wright, Paul Grubbs. Weak Fiat-Shamir Attacks on Modern Proof Systems. *IEEE S&P 2023*. **Distinguished Paper Award**.
2. **Quang Dao**, Paul Grubbs. Spartan and Bulletproofs are simulation-extractable (for free!). *EUROCRYPT 2023*.
1. **Quang Dao**, Julian Wellman, Calvin Yost-Wolff, Sylvester W. Zhang. Rowmotion Orbits of Trapezoid Posets. *The Electronic Journal of Combinatorics*, P2-29, 2022.

Preprints

5. Behzad Abdolmaleki, Matteo Campanelli, **Quang Dao**, Shadman Mohammadi, Nahid Roustaeifar. Resettable Non-Interactive Zero-Knowledge: Attacks and Defenses. *ePrint 2026*.

4. Devon Tuma*, **Quang Dao***, James Waters, Alexander Hicks, Nicholas Hopper. VCVio: Verified Cryptography in Lean via Oracle Effects and Handlers. *ePrint 2026*.
3. **Quang Dao***, Ari Biswas, Liam Eagen, Andrew Milson, Shahar Papini, Justin Thaler. The Sum-Check Protocol over the Monomial Basis, and Other Optimizations. *ePrint 2026*.
2. **Quang Dao***, Zachary DeStefano*, Suyash Bagad, Yuval Domb, Justin Thaler. Speeding Up Sum-Check Proving (Extended Version). *ePrint 2026*. (Combines and improves upon results from *ePrint 2025/1117*, *2024/1046*, and *2024/1210*.)
1. **Quang Dao**, Justin Thaler. Constraint-Packing and the Sum-Check Protocol over Binary Tower Fields. *ePrint 2024*.

Projects

- ArkLib** github.com/Verified-zkEVM/ArkLib. Lean library for formally verified SNARKs. Founder and co-lead developer.
- VCVio** github.com/Verified-zkEVM/VCVio. Lean library for machine-checked cryptographic proofs; ArkLib is built on top of it. Co-lead developer.
- PolyFun** github.com/Verified-zkEVM/PolyFun. Lean library for polynomial functors, interaction trees, and generic interaction frameworks. Spun out of VCVio and used by VCVio and ArkLib. Founder and co-lead developer.
- Jolt** github.com/a16z/jolt. State-of-the-art zero-knowledge virtual machine (zkVM) based on structured lookup arguments. Main contributor.

Grants & Fellowships

- 2025 **Ethereum Foundation Grant**
Grant for the formalization of the FRI-Binius Polynomial Commitment Scheme
Amount: \$24,000
- 2025 **Ethereum Foundation Grant**
Grant for the development of ArkLib, a Lean library for formally verifying SNARKs
Amount: \$100,000
- 2024-2025 **Quad Fellowship**
One-year fellowship for US-based PhD students in STEM fields
Amount: \$40,000
- 2024-2025 **CyLab Fellowship**
One-year fellowship from CyLab at Carnegie Mellon University
Amount: \$50,000

Honors & Awards

- 2024 **Best Paper from Early Career Researchers**
Awarded to the best paper from early career researchers at CRYPTO 2024
- 2023 **Distinguished Paper Award**
Awarded to top 6% of accepted papers at IEEE Security & Privacy 2023
- 2020 **Russell C. Mills Award**
Awarded to 2 seniors for excellence in computer science at Columbia
- 2017 - 2019 **Van Amringe Math Prize**
Awarded annually to the top 3 non-senior students in math at Columbia

2016, 2018 **Putnam Math Competition.** Honorable Mention (top 50)
2016 **International Math Olympiad.** Silver Medal

Industry Experience

Jan 2026 – Present Part-Time Researcher at LayerZero.
Summer 2025 Part-Time Contractor with a16z crypto.
Summer 2024 Research Intern at a16z crypto.

Talks

9. Evolving the Foundations of ArkLib
 - ZKProof 8, Rome (May 2026)
8. ArkLib: Compositional Verification of SNARKs
 - Real World Crypto (Mar 2026)
 - Verified zk(E)VM Seminar (Jan 2026)
 - King's College London (Oct 2025)
 - University of Cambridge (Oct 2025)
 - Ethereum's PQ Interop (Oct 2025)
 - CMU PLunch (Sep 2025)
 - University of Luxembourg Crypto Seminar (July 2025)
 - CRYPTO Day, Telecom Paris (June 2025)
 - EPFL Crypto Seminar (June 2025)
 - CMU Formal Cookies (Oct 2024)
7. Lossy Cryptography from Code-Based Assumptions
 - MIT CIS Seminar (Nov 2024)
 - BUsec Seminar (Nov 2024)
 - NYU Crypto Reading Group (Nov 2024)
 - CMU CyLab Crypto Seminar (Sep 2024)
 - UCLA Crypto Reading Group (Apr 2024)
 - UToronto Theory Seminar (Mar 2024)
6. Non-Interactive Zero-Knowledge from LPN and MQ
 - CMU CyLab Crypto Seminar (Sep 2024)
5. Advanced Security for SNARKs: A Survey
 - a16z Crypto Research Seminar (Jun 2024)
4. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN
 - JP Morgan AlgoCRYPT Seminar (Dec 2023)
 - CMU Crypto Seminar (Nov 2023)
 - NTT Research Seminar (Oct 2023)
 - CyLab Partners Conference (Oct 2023)
 - Vietnam Mathematical Congress (Aug 2023)

3. Weak Fiat-Shamir Attacks on Modern Proof Systems
 - Real World Crypto (Mar 2024)
 - CMU CyLab Security Seminar (Nov 2023)
 - Cornell Security Seminar (Sep 2023)
 - NYU Crypto Reading Group (Sep 2023)
 - Workshop on Attacks in Cryptography (Aug 2023)
 - IEEE S&P (May 2023)
2. Spartan and Bulletproofs are simulation-extractable (for free!)
 - Stanford Crypto Reading Group (May 2023)
 - Telecom Paris Seminar (May 2023)
 - Lattices Meet Hashes Workshop, EPFL (May 2023)
 - CMU Crypto Seminar (April 2023)

Service

Program Committee ASIACRYPT 2026

External Reviewer 2026: CRYPTO
 2025: EUROCRYPT, IEEE S&P, CRYPTO, RANDOM, TCC
 2024: STOC, EUROCRYPT, TCC
 2023: ASIACRYPT, TCC, FOCS
 2022: CRYPTO, EUROCRYPT

Co-Organizer 2022-2025: CMU Crypto Seminar

Teaching

2023-2025 **Western Pennsylvania ARML Team**, *Assistant Coach*, CMU

Fall 2023 **Undergraduate Quantum Computation**, *Teaching Assistant*, CMU

2020-2021 **Calculus I**, *Lead Instructor*, University of Michigan

2017, 2021 **Math & Science Summer Program (MASSP)**, *Mentor*, Vietnam

Mentorship

Chung Nguyen Post-baccalaureate, 2025–Present

James Waters Undergraduate, Fall 2025–Summer 2026

Liam Schilling Undergraduate, Spring & Fall 2025

Ian Martin Undergraduate, Summer 2025

Jack Zhu Undergraduate, Spring 2025

Christian Ang Undergraduate, Spring 2025

Miscellaneous

Languages English (fluent), Vietnamese (native), French (elementary), Malayalam (elementary)

Other writing **Quang Dao**, Christina Meng, Julian Wellman, Zixuan Xu, Calvin Yost-Wolff, Teresa Yu. Extended Nestohedra and their Face Numbers. *arXiv 2019*.